Project Information			
Group ID	30		**Semester**
1. Student Name	Georgios Sandalidis		1
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Project Title	ThreadChat: A Replicated Threaded Group Chat		

Project Description	ThreadChat is a group chat for the command line, written in Python. Users choose a username, join channels and talk to everyone in them. Any message can start a thread: others reply to it, and the replies appear indented under the message they answer. Users can react to messages, edit or delete their own, see who is online and look back through the chat history. The chat does not run on one central server but on a small group of three to five chat servers, usually on different laptops. Every server keeps a full copy of the chat, so no single machine is critical. Nobody types in an address: clients and servers find each other on the local network by themselves, and a new server can join in the middle of a conversation and receive the full history. ThreadChat promises its users three things: - Nothing gets lost. If the server a user is connected to crashes, the client switches to another server automatically and the conversation continues, with no missing or duplicated messages. - Replies make sense. A reply never shows up before the message it answers, even when the two users are connected to different servers. - Everyone sees the same conversation. All servers end up with the same messages, even when the network drops some of them along the way. Project requirements: the servers form a peer group that clients connect to (hybrid architecture). Servers and clients find each other by broadcast (dynamic discovery). The servers elect a leader that decides which servers belong to the group (leader election). Crashed servers, including the leader, and crashed clients are detected and handled (fault tolerance). Messages reach all servers in causal order, so replies always come after the message they answer (reliable ordered multicast).

System Architecture Desin
Include a clear diagram of your system architecture (key components and connections between them).